

Moving Embodied Design Education Online.

Experiences from a Course in Embodied Interaction during the COVID-19 Pandemic

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ABSTRACT

During the rapid and forced move to online teaching during the Covid-19 pandemic, university courses that never would have been considered suitable for online teaching under normal circumstances, were moved online. While challenging, this also opened opportunities for developing innovative strategies for teaching and learning. We report on the experiences of moving online a course in embodied interaction, which due to its focus on situated and embodied design faced extensive challenges in moving online. We highlight in particular the way in which the course brought forward innovative methods for bodystorming and user-assisted trialling. This paper has been written jointly by students and teachers from the course.

CCS CONCEPTS

• **Human-centered computing** → Interaction design process and methods; Applied computing; Distance learning.

KEYWORDS

Design Education, Online Education, Covid-19, Embodied Interaction

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1 INTRODUCTION

During the spring of 2020, University education around the globe moved online, with little preparation, due to the Covid-19 pandemic. Youngmeyer [33] discusses the move as a ‘crash course’ in online teaching and learning, and some have even argued that the move online was ‘successful’ (e.g. [1]) and can be taken as a template for how to integrate online resources better into education in the future. This is a claim that can be debated, as distance education has been extensively researched and that best practices have been already articulated before the pandemic.

However, one of the effects of the rapid and forced move to online teaching was that courses that seldom would have been considered suitable for online teaching under normal circumstances, also were moved online. In this article, we report on one such example: a course in embodied interaction design that incorporates a design project which requires engagement both with physical design materials as well as embodied design exercises. The questions we pose in this article are *How can embodied design education be moved online? What are the challenges, and what are the opportunities offered by the online situation?*

We report on the strategies that made the move possible for us, and elaborate on what worked and what did not work. This article was co-produced by teachers and students from the course, and draws upon our individual reflections as well as student assessments during and after the course. We present the challenges that students and teachers faced, and surface strategies and techniques that allowed us to move a highly physical design project online.

2 BACKGROUND

2.1 Design Education and Embodied Interaction

Within HCI, third wave HCI [9] and its focus on ‘pragmatist, experiential and speculative’ design [7] have brought about an emphasis on interaction design and designerly skills [2]. This has affected educational models to adopt studio-style methods [22] for teaching and learning, with a focus on design critique [10, 21] as its signature pedagogy [23]. In design critique, teachers and peer students engage with each other’s design sketches

to scrutinize their intent as well as their emerging form, with the purpose of coaching students in developing their own judgement and problem-solving skills [10]. Rivard and Faste [18] argue that design education must resist teaching set structures and schema for design, and instead focus on challenging students with unobvious exercises and providing frequent and fluent feedback.

As originally articulated by Dourish [4], Embodied Interaction presents a phenomenological perspective providing a philosophical and methodological grounding to, at the time, emerging trends in interaction research (most notably tangible interaction and social computing). In developing a designerly perspective towards embodied interaction, some rather unusual practical skills need to be developed. One is the need to learn how to design for, but also with, the body and bodily experiences [11, 13, 29]. Educational activities are needed in which students use their own bodies as resources in design exploration [14], as well as learn ways of documenting body and movement-based activities and experiences [31]. A second central skill is the exploration of a wide range of design materials. The technical [28], but also physical and sensory [30] exploration of materials is key to understanding their aesthetic qualities, what they can do, and what they can become. These are skills that are well suited for a physical studio format [12].

2.2 Best Practice for Online Education

Online education is “institution-based, formal education where the learning group is separated, and where interactive telecommunications systems are used to connect learners, resources, and instructors” [25]. While online education was conceived as early as in the seventies [27], it has become an increasingly common practice during the last decades [24], supported by the maturing of tools and technologies. In a literature review from 2011, Simonson et al. [24] come to the conclusion that distance education can be as effective as classroom education, and equally appreciated by students; also that the successful distance learner is an ‘intrinsically motivated’ and ‘abstract’ learner. However, what is significant for such studies is that they have been performed with courses that were 1) partially or fully designed to be taught online, and 2) enrolled students knew in advance that the course would be a distance course.

3 CASE: ONLINE EMBODIED INTERACTION

Drawing a highly international group of students, the two-years HCI master program at Uppsala University is more than ten years old. The course in Embodied Interaction is a ten-week long full-time (40 hours/week) course, offered as an elective course. It comes after courses introducing the students to HCI, user research, scientific methods, user-centered design and interface programming. The course foregrounds theories and methods for interaction design focused on the bodily, social and spatial context of interaction.

A unique feature of the course within the program is the design project’s highly physical nature, especially in how it focuses on designing physical prototypes for strongly situated contexts. Each year, the project revolves around a design theme proposed by the teachers. Students form project groups (3 to 5 people), and each group engages in several phases during which they develop their

design solutions. First, students investigate a domain of practice of their own choice. They engage in ethnomethodological investigations exploring their selected practice, leading them to identify a particular activity or situation that offers opportunities for designing an interactive intervention. Next, students enter the ideation phase for the identified design opportunity. They are therefore introduced to, and engage in, embodied design methods [32] such as bodystorming [16, 20], use-case theater [17], sensitizing [30] and estrangement techniques [32]. In the third and final phase, lasting about 4 weeks, each group engages in iterative design and trial of prototypes that address the identified design opportunity. Students have access to digital and non-digital materials, as well as receive support and elective time to develop and refine their prototypes with the help of the teachers, including dedicated prototyping workshops. They also receive basic instruction in physical prototyping using Arduinos, sensors and actuators, through a series of physical programming labs. Throughout this phase, students engage in design critique seminars facilitated by the teachers, where they practice giving and receiving constructive critique. Lastly, they are required to routinely test their prototypes with target users; as authentically as possible.

Due to the Covid19 pandemic, the Spring 2020 edition of this course had to be re-planned and moved online in its entirety. All of the activities discussed above needed to be moved online, not just lectures and seminars but also the actual design work.

4 CHANGES AND EXPERIENCES

In this section, we summarize the most salient methods and associated experiences. The report below builds on the individual experiences of the authors (who represent both teachers and students), course evaluation reports performed during and after the course, and an in-depth discussion among the authors.

4.1 Course Theme

The first, and critical, change for the course related to how the project theme was selected. Previous years, themes have been focussed either on physical practices (e.g. outdoor training) or related to physical environments (e.g. urban ecology). As the course could no longer require that students observe spaces or activities on-site, it needed to re-focus on human practices that already are to some extent mediated through online communication. The theme chosen for this year was “Social connectedness in times of social isolation”. Within this broad thematic, student groups chose their own focus, subject to approval by the teachers.

In addition to making observations and testing possible, the theme was considered timely and relevant by students, which added to their engagement with the course. One of the reported reasons why the theme was appreciated was that the students themselves were experiencing social isolation first-hand. Choosing a timely theme also meant that students were able to target situations that arose precisely due to the pandemic. The groups ended up choosing the following themes: online groupwork (5 students), social gaming (5 students), parent – children long-distance interaction (5 students), distance romantic relations (3 students) and social eating (5 students).

4.2 Domain Ethnography and Analysis

The extensive domain analysis phase of the course has been developed to counter a tendency among students to primarily focus on screen-based interaction design. The intended outcome of this phase is that students are able to develop their design goals in a situated manner, reflecting on existing practices in their domain from a situated and (physical as well as social) embodied perspective.

The approach proved to work well also in the online setting, even though the theme of the year could potentially have made this tendency even stronger. Students engaged in ethnographic observation of online activities both physically and online. The group working with long-distance romantic relations was located in a country with low virus spread, and therefore able to visit their participants physically in their homes. Other groups found it useful to focus on autoethnographic perspectives, ask participants to write diaries (inspired by Cultural Probes [6]) and conduct observations over videoconferencing tools. The choice of methods was guided by principles of ethnomethodology [5] as they served to uncover everyday practices and norms, and allow for a situated perspective, taking physical and social aspects into account.

4.3 Online Bodystorming

At the start of the design phase, students are introduced to a variety of embodied design ideation methods [31, 32]. The purpose of this is to develop their understanding of embodied and first-person perspectives on design, develop their skills in social and collective designing, and introduce them to the way design ‘talks back’ to designers [22].

After reading about a variety of methods, students are tasked with designing and running a bodystorming [27] session. Bodystorming ideation is typically done with the own body, the space, and various design materials as resources; which was difficult to do as participants were no longer in a particular place or even in the same location at all. In order to develop an understanding of how this could be done online, the course did a small experiment a week in advance, in which each design group planned and executed a very brief bodystorming exercise over Zoom. This session informed their subsequent work with a longer session within their own domain, in which the teachers and at least one of the other student groups would participate. The full-length bodystorming sessions were planned for 45 minutes, the same time span as in previous courses when these sessions were physical. This time span worked well for most groups and was not considered too lengthy.

Sound-Based Instructions versus On-Screen Instructions.

To allow participants to move away from the screen, two groups decided to let one person read instructions out loud. We were for example instructed to lay down on the floor, close our eyes, in exercises inspired by embodied experience practices [30]. In contrast, one group wished to minimize interaction, as a way to invoke a sense of loneliness in the participants. They decided to keep their session quiet and impersonal, posting textual instructions on the screen. Their participants acknowledged that the session was successful in instilling a sense of loneliness, creating a sensitizing experience [26], but was not conducive of ideation.

Facilitation. All groups ended up with one person at a time acting as the facilitator. While bodystorming in general benefits

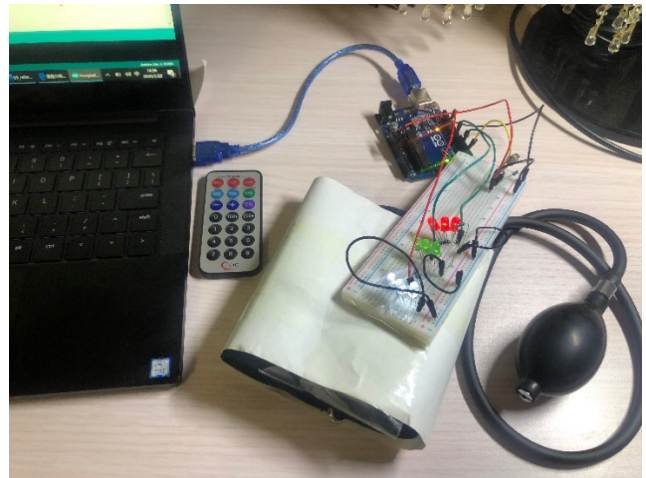


Figure 1: Working with what you have. This picture was taken by one of the students in their home. A blood pressure monitor has been scavenged to create a prototype augmented with interactive light. The prototype simulates a physical “grab” device, for use by romantic couples watching a scary movie together when separated.

from facilitation [27], we have seen this role shift more fluently between members in previous years. While some groups split participants up for debrief and reflection, all groups chose to work with all participants present in the ideation exercises. Having a well-prepared facilitator became even more central than in other bodystorming activities [27]. The facilitator gave instructions to explore materials or move around, among others, and was crucial in reminding of design possibilities beyond the screen.

Challenges in online bodystorming related to difficulties in observing participants, as they sometimes would move away from the camera, as well as noting down their ideas and reflections. Several groups therefore decided to not just rely on their own observations, but to also collect written (and hand-drawn) input from the participants throughout their sessions. These notes also allowed participants to convey a private and first-person perspective on their ideas.

4.4 Materials and Prototyping

In previous years, the course has been highly dependent on students having access to a design lab space, stocked with material such as sticky notes, pens, paper, cardboard, cloth and thread, as well as Arduinos, wires, soldering irons, multimeters etc. This year, kits containing Arduinos, sensors, actuators and a variety of other electronic components were instead assembled and distributed to students. Four online workshops were held, during which students were instructed in using the provided materials and supported in ways to integrate them in their own design work. The kits were based on the TinkerKit Arduino shields and breakout boards.

A make-do attitude. Students developed a ‘make-do’ attitude towards materials. Most students based their designs on materials available in their own place, or that of their test subjects. One example of a student project is depicted in Figure 1. This project is

an example of a wearables project; several students chose to work with wearables as a way to enhance the physical experience of affective communication over distance. Another common approach was to enhance everyday household objects such as cups, chairs and blankets with communicative functions.

Early physical prototypes were largely non-functional, but in later stages these materials were often used together with the Arduino kits. One effect of this 'make-do' attitude towards prototyping was that prototypes would very often only exist in one singular copy, creating challenges for testing them as communicative devices. Students prioritized making prototype output partially functional, mostly through exhibiting some light or sound feedback using 'Wizard of Oz' [3] style of simulations where they would manually trigger the desired feedback in appropriate situations. For example, one student used a heated blanket as a device to represent physical closeness over distance. To simulate its function, the blanket was manually switched when interacted with from another location. Other technology enhanced prototypes included a mug that lit up when used, and a squeezable lamp made from paper and Arduino-controlled LEDs. It should be noted that using interactive technology was not a course requirement, and some prototypes were never developed to a state where they would include electronics.

Tapping into the Home Environment. The fact that each participant would be in their own home was an important resource, in that it provided personally meaningful materials and familiar spaces. The home space became incorporated into the ideation session for most of the groups, and mundane and common home objects (e.g. cups, picture frames, flowerpots, snacks, etc.) became relevant as design materials. Students also brought up how the home environment also became a resource from an ethno methodological perspective. Working from home meant that students were, in a way, 'constantly in the field'. They were able to reflect more on their own lives, something that helped them learn to weave technologies into the fabric of daily life during their design processes.

Working from home however also comes with a range of privacy challenges. When meetings were done from home and designs were done using materials that the group members had access to in their homes, it was difficult for group members to regulate how intimate they wanted to be in their group. Balancing between the need to make the group a working unit and respecting the members' privacy, thus became a challenge for the students.

4.5 Design Critique, Trials and Demonstrations

Sharing design exemplars among students, students and teachers, and even between students and their testers, became a challenge. There were multiple reasons for this, but the most salient one was that they were early prototypes which, whether functional or not, could not be tested without hands-on instructions from their designers. Among the participant groups, there was only one for which it was possible – and sufficiently safe – to visit their participants in person.

4.5.1 Experiences. While some students resorted to telling their test participants what the prototypes would do, rather than letting them test the prototypes in action, several groups found ways to instead engage their participants more deeply in the trial process.

Very often, trial participants would not just be testing a given design, but tasked with building it according to instructions from the student designer.

We saw two versions of this. One of the groups, interested in social eating, recruited their course mates and instructed them in building their own (slightly simplified) versions of the design using the distributed Arduino kits and household items they had at hand. Another group, focusing on distance relations between parents and adult children, recruited their own parents and instructed them in the construction of non-functional prototypes made from simple materials and household items (e.g. a coin tied on a string, representing a medal).

The active involvement in constructing prototypes had an interesting effect, in that trial participants were introduced both to the intended purpose of the design, and the technical and structural aspects of the prototype. We believe that this level of insight allowed them to articulate their design critique in a more precise way. For example, students experienced instances when their participants would *not* build the exact same thing as instructed, instead arguing for a change in the design.

For trialling, one method that worked particularly well was a kind of '**distributed use case scenarios**' setup. These were small, distributed theatre performances wherein the group members enacted a usage situation by taking different roles in a distributed setting. The use of videoconferencing allowed students to choose their locations carefully, and even receive help from friends and family in enacting their designs. One person would typically act as narrator/facilitator to make the situation clear for the audience. This was typically one of the students, but on one occasion, a family member had been recruited to this role, holding the camera and instructing the other members of the family.

5 SYNCHRONOUS ONLINE VERSUS DISTANCE EDUCATION

Courses moving online during the pandemic bore little resemblance with best practices for distance education. In many cases, the online move rather became an exercise in 'online campus education', in adapting existing structures to the online situation with minimal changes. An example of this is how videoconferencing tools became the go-to solution for most educational activities, used for all of lectures, seminars and tutoring sessions. The course reported in this paper is no exception, in particular in how it remained largely synchronous. The design exercises, the trials and the demonstrations detailed in this paper were all done in joint online sessions. The major exception was teacher feedback on texts and documentation, which was asynchronously given through written comments. Since students had signed up for a full-time campus course, the synchronous schedule did not create extensive problems (except for students that had chosen to return to their home countries and were in a different time zone).

We believe that the synchronous nature of our course is not unique, and that many of the courses moving online during the 2020 pandemic employed similar strategies. In particular, courses moving online seem to have heavily relied on video conferencing tools [1, 19]. But this is not in line with current best practices for distance education. One of the key advantages of distance education

is how it allows students to engage in learning activities according to their own schedule and availability. Students who actively choose to engage in distance education do so because they have other engagements, such as working part- or full-time [8].

It is difficult to see how studio-based online design education can become fully asynchronous. Even experimental platforms (e.g. [15]) focus primarily on supporting distributed but not asynchronous education. In our case, bodystorming, trials and demonstrations were all run synchronously. We also found that the peer design critique worked best when it was synchronous, such as within groups and during final demonstrations, and enabled students to be creative and suggestive of new design ideas.

6 CONCLUSION

The move of the Embodied Interaction course online was rapid and rushed, an emergency solution rather than a new form of design education. Some of the reorganizations, such as the change of theme, worked very well. We have highlighted in particular the innovative creative practices of online bodystorming and user-assisted trialling, that emerged as a result of the move and provided opportunities for developing new teaching and learning practices for embodied design in online settings. Giving students great flexibility in developing their own projects and ways of working was instrumental for the course to be carried out online, and we have seen how this also led to very creative ways of working. Teachers and students collaborated in inventing new methods for brainstorming, testing and demonstrations, in order to adapt to the online format.

Apart from the obvious challenges related to the physical nature of course exercises, we have discussed how maintaining a level of privacy became a challenge, in particular in the context of group work. We also want to bring up that allowing for great flexibility added to the already high workload in the course.

The methodological innovations discussed in this article thus came at a cost. It is our hope that our experience can provide useful insights for future educational development, easing the path towards more deliberate and developed strategies of online and blended education that includes physical and embodied design activities, than were available in the spring of 2020.

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